Exercise 1 :

Consider the state space from Figure 1.

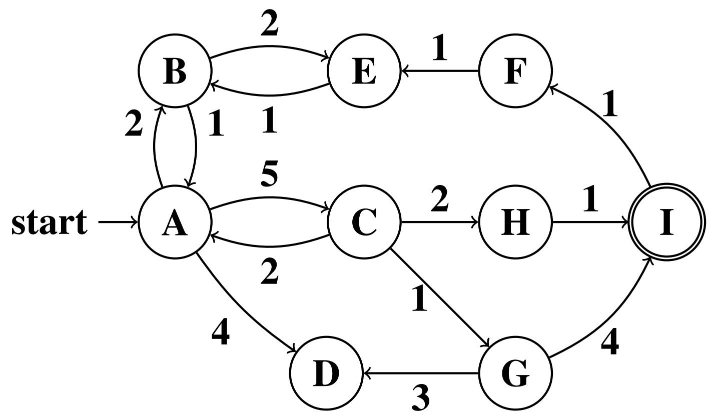


Figure 1: State space.

As a heuristic estimate for a state s, use the minimal number of edges that are needed to reach a goal state from s (or ∞ if s is not solvable, e.g., $h(C) = 2$).

As tie-breaking criteria, if the choice of the next state to be expanded is not unique, expand the lexicographically smallest state first (so A would be expanded before B if both have the same f-value).

(i) Run Uniform-cost-search on this problem. Draw the search graph and **annotate each node with the g value as well as the order of expansion**. Draw duplicate nodes as well, and mark them accordingly by crossing them out. Give the solution found. Is this solution optimal? Justify your answer.

(ii) Run A^* search on this problem. Draw the search graph and **annotate each node with the g and h value as well as the order of expansion**. Draw duplicate nodes as well, and mark them accordingly by crossing them out. Give the solution found by A^* search. Is this solution optimal? Justify your answer.

(iii) What is the minimum number of states that must be expanded by uniform-cost-search regardless of tie-breaking?

(iv) What is the lower bound on the minimum number of states that must be expanded by A*search regardless of tie-breaking (ignoring all states with f equal to the optimal solution cost).

(v) In terms of runtime, which algorithm will find the optimal solution faster. Justify your answer.

Exercise 2 :

Formalize the following problems as state-space problems: define a suitable state space, a start state, set of actions, the action function, and a goal test.

a. In the movie *Die hard: With a vengeance*, Bruce Willis and Samuel L. Jackson are given a water jug riddle, which they need to solve in order to disarm a bomb: There is a water fountain and two jugs that can hold 3 and 5 liters of water, respectively. How do you measure up 4 liters of water (to disarm the bomb) using only the two jugs?

b. (From Russel & Norvig, Exercise 3.9): The *missionaries and cannibals problem* is usually stated as follows: Three missionaries and three cannibals are on one side of a river, along with a boat that can hold either one or two people. Find a way to get everyone to the other side of the river without ever leaving a group of missionaries in one place outnumbered by the cannibals in that place.

Note: This exercise is for you to reflect in the problem definition. Do not spend too much time "formalizing" the problem if you find it difficult, just provide an intuition of how these problems can be encoded as state space search problems.

Exercise 3 :

Consider a variant of the Vacuum Cleaner problem from the lecture where a robot has to clean a 3×3 square room (see Figure 4). There are four possible actions: up, left, right, and down. There is no suck action since the robot automatically cleans any dirty spot it stands on. Hence, its task is moving around the room and visit all dirty spots. Throughout the exercise, we use Manhattan distance.

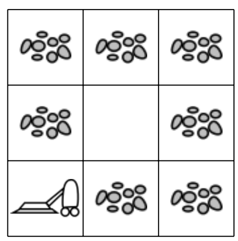


Figure 4: Illustration of the Vacuum Cleaner problem

Which of the following heuristics are admissible? Why / Why not? Give clear proof arguments. (Formal proofs are not needed.)

(i) $h_1 =$ Number of dirty spots.

(ii) $h_2 =$ Number of clean spots.

(iii) $h_3 =$ Sum of the distances from the robot to all the dirty spots.

(iv) $h_4 = h_1 + h_2$.

(v) $h_5 =$ Minimum distance from the robot to any dirty spot.

Note: To prove a heuristic admissible, you need to show that it is admissible for every state of the illustrated 3×3 example. To show that a heuristic is not admissible, a counter example is sufficient.

Exercise 4 :

Consider a robot that can move in the grid below. The cost of moving from one cell to another is equal to the number of steps required for the move, where by a single step the robot can move horizontally or vertically to an adjacent cell (no diagonal steps allowed).

				S			
		X_1				X_5	
	X_2				X_4		
			X_3				

The cells marked X_1, X_2, X_3, X_4, and X_5 are *cells of interest*, each holding one or two of the symbols a, b, and c:

- X_1: (a, c)
- X_2: (b)
- X_3: (b, c)
- X_4: (c)
- X_5: (a)

That is, X_1 holds the symbols a and c. When visiting a cell the robot collects the symbols located at that cell.

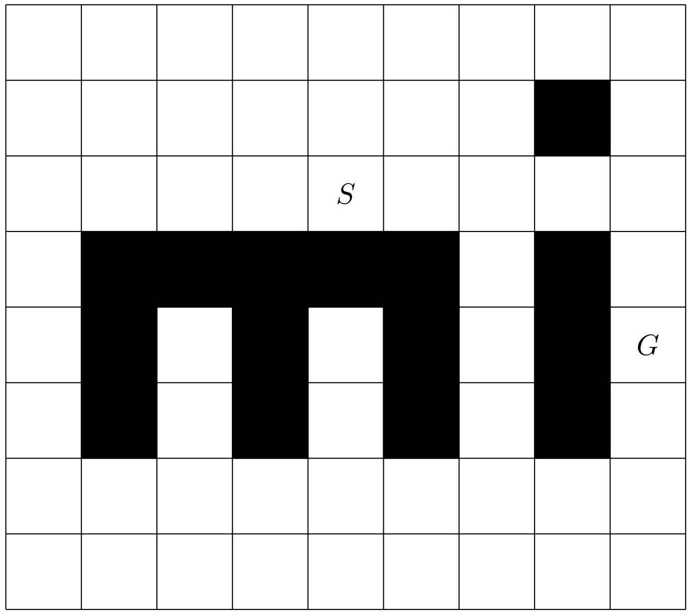
Starting in cell S, the robot's task is to visit cells of interest and collect exactly one copy of each of the three symbols (a, b, c) before returning to cell S. The robot is *not* allowed to move to a cell containing, e.g., an a if the robot has previously visited a cell holding an a.

Determine which of the cells of interest to visit and in which order these cells should be visited by the robot, so that the cost of the path traveled (when starting and ending in cell S) is minimized.

1. Define a suitable state and action representation for this problem.
2. Define a heuristic function that at any given state provides an underestimate of the cost of reaching a goal state (i.e., a state where the robot is located in cell S with the symbols (a, b, c)).
3. Show the search tree for this problem as it would be expanded using A* search.

Exercise 5 :

Consider the problem of finding a path from position S to G in the grid shown below. You can only move horizontally and vertically and only one step at a time; no step can be made into the shaded areas or outside the grid. The cost of the path between two positions is the number of steps on the path.



1. Number the positions (cells in the grid) in the order in which they are added to the frontier in a uniform-cost search for G starting at S. Assume that the ordering of the operators is *right*, *down*, *left*, and *up*, and that there is multiple path pruning. When multiple lowest-cost paths exists, choose the path first added to the frontier.
2. Define a heuristic function that underestimates the cost of the lowest-cost path.
3. Number the positions (cells in the grid) in the order in which they are added to the frontier according to an A^* search using your heuristic function. Resolve ambiguities using the operator ordering and the procedure above.

NB: When numbering the positions it may be helpful to also annotate the positions with the cost/function values calculated during the search.
